

GVPT Data Science Camp

Introduction and Setup

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Today's Plan

- ▶ Welcome and introductions
- ▶ What is R, and why do we use it?
- ▶ Installing R and RStudio
- ▶ A tour of RStudio
- ▶ AI, LLMs, and agentic coding

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- ▶ Please ask questions about **anything**.

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- ▶ Fun fact: I used to operate a college basketball burner Twitter account

Learning Objectives

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- ▶ Getting to know your cohort
- ▶ Introduction to R and RStudio
- ▶ Brief discussion of AI, LLMs, and agentic coding

What is R and why do we use it?

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- ▶ R is a free, open-source programming language designed for statistical computing, data analysis, and data visualization
- ▶ Over the past four decades, quantitative research has dominated social science, and R provides us the ability to conduct such research

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Installing R

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- ▶ Locate the file in your downloads folder and double-click it
- ▶ Follow the setup prompts; keep the default settings. Click “Finish” when complete

What is RStudio?

- ▶ RStudio is an **integrated development environment (IDE)** we use to write code

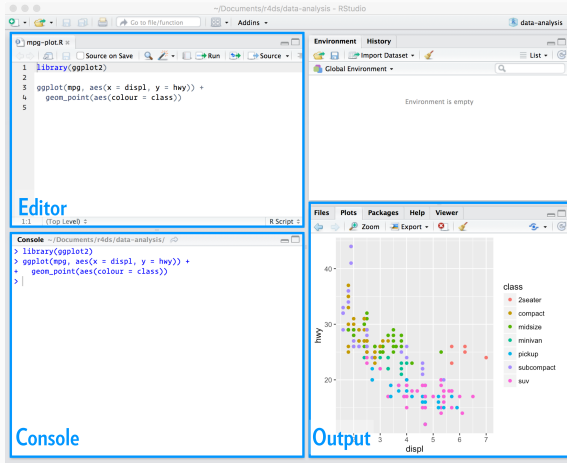
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- ▶ Run the setup:
 - Windows: open the downloaded `.exe` file and follow the setup prompts
 - Mac: open the downloaded `.dmg` disk image and drag the RStudio icon into your Applications folder

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- ▶ Click “Finish” or exit the installer once the process is complete

A Tour of RStudio



Source: *R for Data Science* (R4DS)

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- ▶ **LLMs (Large Language Models)** are a type of AI that are trained on massive datasets to produce human-like text
- ▶ **Agentic coding** is a software development approach where autonomous AI agents plan, write, and test code with minimal human intervention

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- ▶ Brief introduction to the exciting world of AI

Questions?